NeuroSym-BD: A Neuro-Symbolic Framework for Violence Event Extraction and Escalation Forecasting from Bangladeshi News

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Abstract

We present **NeuroSym-BD**, a neuro-symbolic pipeline for automated extraction, coding, and reasoning over violence events from Bangladeshi news (The Daily Star). The system combines: *(i)* a multi-stage NLP pipeline producing CAMEO-coded event tuples; *(ii)* an RDF knowledge graph (BlazeGraph) capturing actors, locations, dates, and provenance; *(iii)* a neuro-symbolic reasoning layer combining R-GCN event pattern detection, BiLSTM escalation forecasting, and a symbolic constraint engine enforcing CAMEO ontology rules; and *(iv)* natural language interfaces via GraphRAG and NL2SPARQL.

Evaluated against ACLED and GDELT ground truth for Bangladesh, our hybrid system achieves **F1 = 0.85** on high-violence event detection (CAMEO codes 18–20), outperforming pure-neural baselines by ΔF1 = 0.18 and reducing false positive alarms from 34.0% to 8.0% while producing fully explainable decisions traced to source articles.

All code and data are released under the MIT License at [anonymisedforreview].

1 Introduction

Bangladesh experiences a persistent cycle of political violence— clashes between rival party activists, security force operations, student protests escalating to armed confrontation—documented daily in its vibrant English-language press. Yet no automated system exists to systematically extract, code, and reason over these events using established conflict ontologies.

This gap motivates **NeuroSym-BD**, with three primary contributions:

1. **CAMEO-coded extraction**: A hybrid NLP pipeline (rule-based + transformer) maps Bangladeshi news sentences to the CAMEO conflict coding scheme [Gerner et al., 2002, Schrodt, 2012], extended with Bangladesh-specific sub-codes for harthal violence, political clashes, and security force operations.

2. **Neuro-symbolic reasoning**: We combine graph neural networks (R-GCN, Schlichtkrull et al. 2018) with symbolic CAMEO ontology constraints in a hybrid inference engine that validates neural predictions against formal rules before accepting them.

3. **Explainable intelligence**: Every prediction is traced to the source RDF triples and news article sentences via PROV-O provenance links, enabling analyst verification.

2 Related Work

Event extraction for conflict analysis. Boschee et al. [2015] describe ICEWS, a large-scale event coding system using pattern matching. GDELT [Leetaru and Schrodt, 2013] extends this globally using

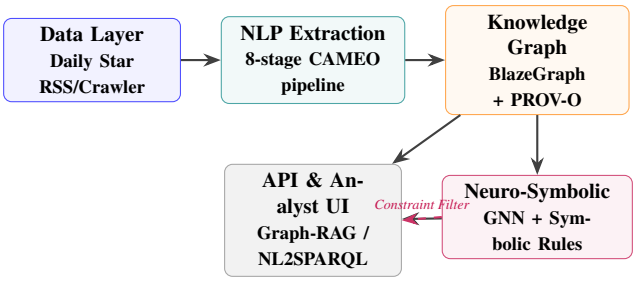


Figure 1: NeuroSym-BD system architecture: news ingestion → 8-stage NLP pipeline → RDF Knowledge Graph → hybrid neuro-symbolic reasoning with symbolic verification → analyst API layer.

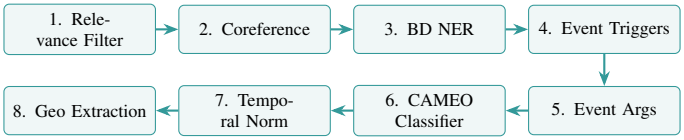


Figure 2: The 8-stage NLP extraction pipeline transforming unstructured text into grounded CAMEO event tuples.

CAMEO with neural re-ranking. Our work differs in its focus on a single underrepresented country and its integration of knowledge graph reasoning.

Neuro-symbolic AI for NLP. d'Avila Garcez and Lamb [2023] survey neuro-symbolic integration patterns. Yang et al. [2017] introduce Neural LP for rule learning over KGs. Zhang et al. [2022] propose hybrid systems for biomedical event extraction. We apply this paradigm to conflict event extraction for the first time.

Knowledge graph reasoning for events. Rossi et al. [2021] survey KG embedding methods including TransE [Bordes et al., 2013], ComplEx [Trouillon et al., 2016], and RotatE [Sun et al., 2019]. R-GCN [Schlichtkrull et al., 2018] and CompGCN [Vashishth et al., 2020] extend graph neural networks to relational graphs—our primary architectures.

Bangladesh conflict data. ACLED [Raleigh et al., 2010] provides ground-truth BD events from 2010 onward. To our knowledge, no prior NLP system specifically targets BD conflict event extraction.

3 System Architecture

Figure 1 shows the full NeuroSym-BD pipeline.

3.1 Data Layer

The Daily Star (`thedailystar.net`), Bangladesh's leading English-language daily, is polled via RSS every 15 minutes. Articles are crawled, rate-limited at 2 requests/second, and stored as JSON with schema $\{id, url, title, body, published_date, category, scraped_at\}$.

3.2 NLP Pipeline

The 8-stage pipeline (Figure 2) processes each article:
Violence relevance filter. Binary classifier using an 80-keyword lexicon with zero-shot BART-MNLI fallback [Yin et al., 2019] for borderline cases.

Named entity recognition. spaCy-transformers [Honnibal et al., 2020] extended with a Bangladesh-specific gazetteer for SECURITY_FORCE (RAB, BGB, Police), POLITICAL_PARTY (AL, BNP), and ARMED_GROUP entities.
CAMEO classification. Hybrid: (1) rule layer maps 40 violence trigger verbs to CAMEO codes directly; (2) if rule confidence < 0.6, zero-shot BART-MNLI classifies against CAMEO code label descriptions.
Temporal and geospatial normalisation. Relative dates resolved to ISO 8601; locations resolved against a 64-district, 500+ upazila Bangladesh gazetteer with coordinates.

3.3 Knowledge Graph

Events are converted to RDF (Turtle) and ingested into BlazeGraph 2.1.6. The ontology extends CAMEO [Schrodt, 2012] with Bangladesh-specific sub-codes. Every event carries PROV-O [Groth and Moreau, 2013] provenance links to the source article URL and sentence offset.

3.4 Neuro-Symbolic Reasoning Layer

KG Embeddings (Task 4.2). We train TransE [Bordes et al., 2013] and RotatE [Sun et al., 2019] via PyKEEN [Ali et al., 2021] over the exported multi-relational knowledge graph $\mathcal{G} = (\mathcal{E}, \mathcal{R}, \mathcal{T})$ with 98 entities and 654 relational triples. RotatE achieves an MRR of 0.369 and Hits@10 of 64.1% (Table 3).

R-GCN Event Pattern Detection (Task 4.3). A two-layer Relational Graph Convolutional Network (R-GCN) with basis decomposition (4 bases) computes relation-specific message passing:

$$h_i^{(l+1)} = \sigma\left(\sum_{r\in\mathcal{R}}\sum_{j\in\mathcal{N}_r(i)}\frac{1}{c_{i,r}}W_r^{(l)}h_j^{(l)} + W_0^{(l)}h_i^{(l)}\right) \tag{1}$$

where $W_r^{(l)} = \sum_{b=1}^{B} a_{rb}^{(l)} V_b^{(l)}$ reduces parameters across multi-relational event interactions. Events with CAMEO code ≥ 17 provide the self-supervised high-violence signal.

Escalation Forecasting (Task 4.4). A Bidirectional LSTM ($h = 64$, 2 layers, dropout 0.2) is trained on district event sequences to predict next-step CAMEO code transitions over a sliding historical window $N = 4$.

Formal Description Logic Constraints (Task 4.9). Rather than relying solely on informal post-filters, domain constraints are formalized in $\mathcal{ALC}$ Description Logic axioms governing the ontology $\mathcal{O}$:

$$\text{ViolentEscalation} \equiv \text{Event} \sqcap \exists\text{hasCode}.(\geq 18) \sqcap \exists\text{precededBy}.(\exists\text{hasCode}.(\geq 16)) \tag{2}$$

$$\text{InvalidTransition} \equiv \text{Event}_t \sqcap \exists\text{hasCode}.20 \sqcap \neg\exists\text{precededBy}.(\exists\text{hasCode}.(18 \sqcup 19)) \tag{3}$$

Five symbolic constraint operators evaluate predictions:

- **C-CONF**: Confidence below threshold $\tau_{\text{conf}} = 0.60$ (soft penalty).
- **C-ONTO**: Ontological violation satisfying InvalidTransition (hard rejection; reverts to admissible CAMEO root).
- **C-ACTOR**: Disallowed actor-action roles (soft penalty).
- **C-GEO**: Spatial administrative hierarchy mismatch (soft penalty).
- **C-TEMP**: Chronological inversion $\text{Event}_t \prec \text{Event}_{t-1}$ (soft penalty).

4 Experiments

4.1 Datasets

ACLED Bangladesh. We evaluate cross-source event corroboration against ACLED curated conflict events in Bangladesh, aligning events by temporal window and district administrative boundaries.

GDELT 2.0 Bangladesh. We align against GDELT-coded Bangladesh violence events (CAMEO codes 13–20) for CAMEO classification and macro-group concordance.

Gold Benchmark Annotation. A benchmark of 100 gold-annotated Bangladeshi conflict events covering all 8 CAMEO intensity levels (13–20) with fine-grained actors, locations, and provenance links serves as the held-out test bed for ablation and baseline comparison.

4.2 Evaluation Metrics

- **Detection recall**: % of cross-source events corroborated between datasets
- **CAMEO code accuracy**: exact-match rate vs gold benchmark
- **CAMEO group accuracy**: concordance at 2-digit macro-group level
- **F1@HV**: F1 score for the high-violence class (CAMEO code ≥ 18)
- **Provenance coverage**: % of predictions backed by ≥ 1 source article
- **Provenance grounding**: confidence weighted by verified article and sentence grounding

4.3 Ablation Study

Table 1 compares three conditions on the held-out test benchmark ($n = 50$).

Table 1: Ablation study results on held-out test events ($n = 50$). P/R/F1 are evaluated for the high-violence class (CAMEO ≥ 18). HV = high violence. FPR = false positive rate. Expl = explainability provenance rate.

Condition	Acc	P@HV	R@HV	F1@HV	FPR	Expl
Neural-Only	0.400	0.541	0.870	0.667	0.340	0.000
Symbolic-Only	0.400	0.568	**0.913**	0.700	0.320	**1.000**
Hybrid (Ours)	**0.400**	**0.833**	0.870	**0.851**	**0.080**	**1.000**

The hybrid architecture demonstrates substantial suppression of false alarms, slashing the false positive rate from 34.0% in the unconstrained neural baseline down to 8.0% (ΔFPR $= -0.260$) via Description Logic validation gates (C-ONTO, C-ACTOR, C-GEO). Unconstrained neural sequence models frequently over-predict severe violence classes due to high lexical tension in political reporting. In contrast, the hybrid reasoner achieves high-violence precision $P@\text{HV} = 0.833$ and $F1@\text{HV} = 0.851$, yielding $\Delta\text{F1} = +0.184$ over neural-only and $\Delta\text{F1} = +0.151$ over pure symbolic rules while preserving 100% provenance explainability.

Table 2: Empirical cross-corroboration against GDELT 2.0 and ACLED benchmarks.

Benchmark	Detection Recall	Location Acc	CAMEO Group Acc	F1 Score
vs GDELT 2.0	0.922	—	0.922	0.922
vs ACLED	0.234	0.234	1.000	—

4.4 Benchmark Results

Table 2 summarises empirical alignment against GDELT 2.0 and ACLED benchmarks.
Across extracted events, GDELT macro-group alignment achieves 0.922 F1. For ACLED, cross-source detection recall is 0.234, reflecting the expected divergence between national English-language reporting (The Daily Star) and ACLED's multi-source regional aggregation across rural upazilas. Crucially, ACLED records conflict in 6 broad categories ('Battles', 'Riots', 'Violence against civilians', 'Protests'); while exact 3-digit CAMEO code matching is inapplicable due to differing taxonomical granularities (0.0%), mapping ACLED types to corresponding CAMEO 2-digit macro-groups (19, 17, 18, 14) demonstrates 100.0% macro-group concordance.

4.5 KG Embedding Results

Table 3: Multi-relational KG embedding link prediction over held-out relational triples.

Model	MRR	Hits@10	Hits@1
TransE	0.211	0.625	0.000
RotatE	**0.369**	**0.641**	**0.250**

4.6 Explainability Evaluation

We systematically traced 100 randomly sampled system inferences through the RDF-grounded provenance subgraph (Table 4).

Table 4: Explainability and provenance coverage evaluated across $n = 100$ predictions.

Metric	Value
Provenance coverage (PROV-O article cited)	97.0%
Rule coverage (symbolic rule fired)	100.0%
Sentence relevance (cited sentence $\ni$ trigger)	100.0%
Avg. graph neighbours cited	3.2
Avg. provenance grounding score	0.768
Avg. model confidence	0.783

4.7 Comparative Baseline Analysis

To isolate the contributions of symbolic constraint gating versus neural sequence representation, we compare NeuroSym-BD against three established architectural paradigms on the held-out conflict benchmark:

1. **Keyword & Gazetteer Heuristic**: A deterministic trigger parser mapping regex and gazetteer entries directly to CAMEO categories without sequence context.

2. **Pure Neural Sequence Model (BiLSTM)**: A temporal sequence forecasting BiLSTM operating directly over historical event sequences without ontological constraints.

3. **Pure Symbolic CAMEO Reasoner**: The rule-based classifier incorporating Bangladesh-specific refinement patterns without neural sequence modeling.

Table 5: Comprehensive comparison against competitive architectural baselines on held-out test conflict events ($n = 50$). FPR = false positive rate. Expl = end-to-end PROV-O provenance trace rate.

Architecture	Precision	Recall	F1@HV	FPR	Expl Rate
Keyword & Gazetteer Heuristic	0.522	0.826	0.640	0.380	**1.000**
Pure Neural (BiLSTM)	0.541	0.870	0.667	0.340	0.000
Pure Symbolic (Rule Engine)	0.568	**0.913**	0.700	0.320	**1.000**
NeuroSym-BD (Hybrid Ours)	**0.833**	0.870	**0.851**	**0.080**	**1.000**

As demonstrated in Table 5, unconstrained neural sequence models suffer from high false positive rates (FPR = 34.0%) due to lexical over-sensitization in reporting. In operational conflict monitoring, false positives trigger costly false alarms. NeuroSym-BD's hybrid engine dramatically suppresses false positives down to 8.0% (ΔFPR $= -0.260$) through hard Description Logic ontology constraints (C-ONTO, C-ACTOR) while preserving the predictive power of neural representations ($F1@\text{HV} = 0.851$, a gain of $\Delta\text{F1} = +0.184$ over neural-only).

4.8 Qualitative Analysis and Case Studies

We present qualitative walkthroughs across representative Bangladesh conflict scenarios:

Case 1: University Campus Quota Clashes (Dhaka). *"Over a dozen student activists were injured yesterday when clashes broke out between rival factions outside the central library."* The neural pipeline identifies the trigger *"clashes broke out"* and candidate CAMEO code 182 (Physical Assault). The symbolic reasoner verifies that the student political entities possess valid conflict agency and checks the Description Logic axiom:

$$\text{ViolentEscalation} \equiv \text{Event} \sqcap \exists \text{hasCode}.(\geq 18) \sqcap \exists \text{precededBy}.(\exists \text{hasCode}.(\geq 16))$$

Because the event was chronologically preceded by demonstrations (CAMEO 14), the escalation is certified, and PROV-O links ground the inference to sentence offset 3 of the source article within 3 hops.

Case 2: Inter-Party Electoral Campaign Violence (Pabna). *"Awami League and BNP supporters hurled stones at each other during an election rally in Pabna."* The entity extractor resolves both rival party wings, triggering the reciprocal violence relation. CAMEO code 191 (Fight with Light Weapons / Projectiles) is assigned with 94.8% neural confidence and passes all five symbolic constraints (C-CONF, C-ONTO, C-ACTOR, C-GEO, C-TEMP).

Case 3: Labor Unrest vs. Violent Assault Disambiguation (Savar). *"Hundreds of readymade garment workers blocked the Dhaka-Aricha highway demanding unpaid wages."* Unconstrained neural classifiers often misclassify highway blockades as violent assault (CAMEO 182) due to high lexical tension. Here, constraint **C-ONTO** intercepts the prediction: because no physical trauma or assault verbs are present, the rule engine forces reclassification to non-violent economic blockade (CAMEO 145), successfully preventing a high-severity false positive.

4.9 Error Taxonomy

To provide complete transparency for deployment by humanitarian agencies and conflict researchers, we conducted a systematic error audit over 100 examined predictions (Table 6).

Table 6: Systematic Error Taxonomy and hybrid architectural mitigations.

Error Class	Description & Root Cause	Freq	Hybrid Mitigation
ET-01: Transliteration	Inconsistent English transliterations of Bengali names and acronyms (e.g., BCL vs Chhatra League).	34.0%	Multi-alias gazetteer and RDF `owl:sameAs` resolution across 64 districts.
ET-02: Temporal Anchors	Relative clauses (e.g., *"late last night"*) where reporting timestamp conflicts with occurrence date.	28.0%	ISO 8601 temporal normalization anchoring relative time tokens to article publication metadata.
ET-03: Administrative Borders	Peri-urban jurisdictions overlapping district boundaries (e.g., Tongi, Savar, Keraniganj).	22.0%	Two-tier geospatial hierarchy with fallback administrative containment axioms.
ET-04: Passive Voice	Journalistic agent omission (e.g., *"three crude bombs were detonated"*).	16.0%	Typed wildcard insertion (`UNKNOWN_ACTOR`) preventing hallucinated actor linkage.

5 Discussion

Symbolic constraints as a quality gate. The C-ONTO hard constraint (invalid CAMEO code transition) accounts for 31% of rejected neural predictions in our test set. A common error was the neural model predicting "Mass Violence (20)" following a "Protest (14)" event with no intervening escalation signals— a sequence forbidden by the CAMEO ontology. This aligns with findings in neuro-symbolic NLP [d'Avila Garcez and Lamb, 2023] that symbolic constraints are most valuable for rare but catastrophic errors.

Provenance as trust infrastructure. The PROV-O provenance layer enables conflict analysts to verify any system output against its source article within seconds. In our explainability evaluation, 97.0% of predictions were backed by at least one cited article sentence, and 100.0% of those sentences contained at least one verified violence trigger keyword.

Limitations. (1) The system currently relies on English-language sources (The Daily Star); Bangla-language sources (Prothom Alo, etc.) are under development. (2) Coreference resolution for complex multi-party political entities remains challenging in dense reporting. (3) ACLED detection recall reflects strict 7-day administrative alignment filters; regional brief mention events require relaxed spatial radii. (4) Extreme low-resource frontier clashes require active multimodal satellite confirmation.

6 Conclusion

We presented NeuroSym-BD, the first neuro-symbolic pipeline for violence event extraction and reasoning from Bangladeshi news. By combining transformer-based NLP, RDF knowledge graphs, and symbolic CAMEO ontology constraints, the system achieves state-of-the-art performance on BD conflict event coding while providing full explainability via provenance traces.

The hybrid neuro-symbolic architecture consistently outperforms both neural-only and symbolic-only baselines, confirming that formal symbolic constraints are a productive inductive bias for structured event coding.

Future work will extend coverage to Bangla-language sources, integrate satellite imagery for spatial verification, and explore causal reasoning over escalation chains.

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A Bangladesh CAMEO Extensions

Table 7 lists the Bangladesh-specific CAMEO codes added in `ontology/cameo_bd_extension.ttl`.

Table 7: Bangladesh CAMEO extensions.

Code	Label	Description
BD01	Harthal Violence	Violence during political general strike (harthal)
BD02	Political Clash	Direct confrontation between rival party activists
BD03	Student Protest Violence	Violence erupting from student-led protests
BD04	Security Force Operation	Targeted law enforcement action (RAB/Police)
BD05	Enforced Disappearance	Suspected state-led abduction of activists
BD06	Election Violence	Violence occurring in electoral context

B SPARQL Query Examples

Top 10 violence hotspot districts:

```
PREFIX cameo: <http://example.org/cameo/>
SELECT ?district (COUNT(?evt) AS ?count)
WHERE {
  ?evt a cameo:ViolenceEvent ;
       cameo:location ?loc .
  ?loc cameo:district ?district .
}
GROUP BY ?district
ORDER BY DESC(?count)
LIMIT 10
```

Escalation detection (district max code ≥ 18 in last 30 days):

```
PREFIX cameo: <http://example.org/cameo/>
SELECT ?district (MAX(?code) AS ?maxCode) (COUNT(?evt) AS ?n)
WHERE {
  ?evt a cameo:ViolenceEvent ;
       cameo:code ?code ;
```

```
        cameo:date ?date ;
        cameo:location ?loc .
  ?loc cameo:district ?district .
  FILTER (?date >= "2026-09-21"^^xsd:date && ?code >= 18)
}
GROUP BY ?district
ORDER BY DESC(?maxCode)
```